MOBILE FORENSICS EXAMINATION STANDARD OPERATION PROCESURE (SOP)

MAY 7, 2020

Version 1.5

Table of Contents

Table of Contents

Document History 0

1. Introduction 2

2. Seizure 2

3. Approved tools 4

4. Acquisition 6

5. Analysis 10

6. Documentation 20

Works Cited 23

Document History

Date	Version	Description/Comments	Authors
02/11/2020	1.0	Initial Creation of the Document with references	Fatema
03/12/2020	1.1	Added sections (seizures, Documentation) to the document	Fatema
4/15/2020	1.2	• Added Acquisition and Analysis sections • Updated Seizures, and references sections	Fatema
4/27/2020	1.3	Updated Acquisition and Analysis sections	Fatema
5/4/2020	1.4	• Added sections to Acquisition and Analysis • Reviewed and edited entire document	Fatema

5/7/2020	1.5	Final Review	Fatema

1. Introduction

1.1 Purpose

The purpose of the Mobile Forensic Examination Standard Operation Procedure is to provide guidance for the Mobile Forensic examiners. These standards cover the handling and reporting of Mobile Forensic examinations.

1.2 Scope

This standard operating policy utilizes collecting, analyzing, and preserving mobile devices and images. This SOP does not apply to cover all supported devices due to the prevalence of iOS 12 and 13.

1.3 Document Updates

This document must be updated upon major changes to the process of examination and reporting.

This SOP needs to be annually reviewed, updated, signed and shared with employees at a minimum, but also needs to be updated upon major changes.

2. Seizure

Correct protocol and procedure must be followed to seize one or more mobile devices(s) to ensure that there is no loss of data, and that the collected evidence is not ‘spoiled” because of contamination or failure to follow protocol. Seizure procedure must comply with the federal and state laws. Illegal searches could lead to evidence being inadmissible to court.

All seized mobile devices must comply with a search warrant or consent form. Exigent Circumstances, where data could be remotely wiped or written over, are a non-issue during mobile device seizure, because of data preservation or network isolation techniques that could bypass those concerns. Therefore, Seizure process must accompany with a Search Warrant and/or a consent Form.

The Search Warrant or Consent form must define:

- what devices can be searched and seized: in detail descriptions such as the manufacturer, serial and model number, color, any scratches or defects and other relevant details to uniquely identify the device.
- what time and physical locations can be searched
- what additional peripherals such cables, removable media, UICC, etc. can be seized.

Attempts to gather information such as the security code, password or gestures needed to gain access to the device should be made when questioning the owner of the device. Validation and verification methods should be applied, such as testing each solution on an identical test device and under known conditions, then documenting the results and process. However, caution should be made to ensure no master reset code or remote wipe is initialized.

State of the device:

Mobile devices may be found on, off or in a damaged state. In these situations, examiners must be cautious and follow the organizational procedure to ensure minimum loss of data. Devices with physical damage may still be useful for data extraction and examination. In these situations, damaged devices should be taken to the lab and attempts to repair the device for examination should be made. In situations where the device is immersed in liquid, battery can be removed to avoid electrical shortening and kept in a safe and sealed container. For life-threatening situations, appropriate specialists should be consulted.

2.1 Data Preservation:

Make sure to follow data preservation methods to maintain data integrity and ensure minimum data loss. Below are some useful steps:

- Network Isolation, such as placing the device in faraday bags to prevent remote wipes or upgrades to be made in the device
- Maintain Battery
- Keep devices off if they are already off.
- Keep devices on if they were on.
- Locking- if password can be obtained from the user, then try it on live, record and document. After making sure it works, let it lock on its own. Disable timeout locks if possible.

2.2 Network Isolation

Network Isolation procedures must be followed in exigent circumstances, where there is a concern for data loss or modifications, such as in situations where data could be remotely wiped or written over. Below are some examples of network isolation protocols:

- Faraday enclosure/bag
- Alternatives to Faraday bag: Arson can / microwave/ foil –less trustworthy
- Airplane mode: Ensure wi-fi and/or Bluetooth is turned off.
- Sim Card Clone
- Remove SIM for GSM only feature phones that do not have Bluetooth, wi-fi etc.

3. Approved tools

Not all tools are approved by the lab. See document titled, "Lab Approved Tools" for a detailed List. Make sure these tools are validated and verified before each acquisition and examination. Use Multiple tools. Do Not reply on a single tool.

Below are the tools listed, for acquisition and analysis, which are approved by the organization:

3.1 Acquisition Tools

Open Source:

- **Santoku**
- **OSAF**

Commercial:

- **Access Data Mobile Phone Examiner Plus (MPE+):** Android, Blackberry, Windows phone, feature phones
- **Belkasoft Acquisition Tools:** Logical images from Android and iOS devices
- **Black Bag Suite:** Logical images from Android and iOS devices
- **Cellebrite UFED:** Android, iOS, Windows, Blackberry, Feature phones
- **Compelson MOBLedit:** Android, iPhone, Blackberry, Symbian, Windows Mobile, Windows Phone, Bada, Meego, Chinese phones and CDMA phones.
- **Elcomsoft suite**
- **Gray Shift**

- **Magnet Forensics**: Android and iOS.
- **Katana Forensics**: Logical images from Android and iOS devices
- **MSAB Suite**
- **Oxygen Forensics Extractor suite:** Android and iOS, cloud, windows phone, feature phone, etc.
- **Paraben, Susteen, Athena, Logic Cube CellDek**
- **Others listed the document, titled "Lab Approved Tools".**

For SIM and UICC imaging:

- **Pro Data Doctor SIM Card Reader**
- **Cellebrite**
- **XRY**
- **Oxygen**
- **Susteen Secure View**
- **Mobile Edit**
- **USIM Detective**
- **SIMiFOR**
- **SIMIS Tool**
- Other non-forensics tools are also available. See document titled, "Lab Approved Tools" for a detailed list of tools approved by the organization.

3.2 Analysis Tools

Open Source

- **Autopsy:** Android Analyzer will parse most Android Devices
- **aLeapp** – Android Parser
- **iLeapp**- iOS parser
- **Andriller**- Android Analysis

Commercial

- **Cellebrite**
- **XRY**
- **Encase**
- **FTK**
- **Gray Key-** iOS devices File system Analysis
- **CAS- iOS** devices File system Analysis

- **Magnet Forensics Axiom**
- **Oxygen Forensic**

4. Acquisition

There are many kinds of mobile devices, with different make/model, different operating systems (OS) and different versions of the same OS. They have different features and capabilities and will need different forensic approaches to preserve the data properly. Below is a general approach to acquiring the data in forensically clean manner

Below is a general approach to acquire the data in forensically clean manner.

Forensic Imaging

- Determine the device is “off”,” on and unlocked”, “on and locked”, or in a “damaged state”
- Determine the collection type: Physical or Logical
- Consider time, tools, and device state, OS and versions, make/model of the device
- Validate with multiple tools to determine whether the software is modifying the device data upon extractions.
- Verify tools, software and device integrity.
- If the device is on, and unlocked, start with a logical acquisition
- If device is locked, start with physical acquisition
- Document and photograph the process
- Create two copies of the original evidence if possible: a best copy, and a working copy.

For Device received in ‘off’ state

- If the device is off, start with physical acquisition, after taking additional pictures with battery cover off, and removed.
- Obtain forensic images of all other media (UICC card, SD cards, etc.) separately, apply write block appropriately.
- Document and photograph the process

For Device received in ‘on’ state

- If the device is on, then determine to do physical, logical or Files system backups depending on the request information needed/requested.
- Do not access battery until after logical analysis

- Do not remove UICC.
- Do a logical acquisition with applicable tool(s). Validate acquisition
- Move on to photography as if device was off
- Obtain forensic images of additional evidence items (UICC, SD card)
- Document the process

For Devices “on and locked” state:

- Smudge attack
- Access to the cloud account, if possible
- Brute-force is also another option, but keep in mind about failed attempts and device wipe.
- Bypass repair tools can be used such as BST dongle.
- JTAG/Chip- off

For Damaged devices

- Photograph the device including damaged areas
- Attempt to acquire the device with commercial tools.
- If the device is damaged, the device may require JTAG/Chip-off.
- photograph and document before and throughout the disassembly process

JTAG Acquisition:

- TAG (Joint Test Action Group) a physical method that involves cabling and connecting to Test Access Ports (TAPs) on the device and using processing instructions to invoke a transfer of the raw data stored in memory. Raw data is pulled directly from the connected device using a special JTAG cable.
- JTAG acquisition can often be done for locked, damaged and inaccessible (locked) devices.
- After JTAG data is acquired from the device, it can be analyzed with open source or commercial tools

Chip-off acquisition

- Chip off is the removal of memory chips using highly specialized tools to remove the chips and other specialized devices to read the chips from the device, used when device is damaged there is no other way to get to the data

- It produces raw binary dumps
- If chip is not encrypted, a forensic copy can be extracted as a raw image and can be analyzed by open source or commercial tools
- Encrypted data will not be decrypted after chip-off

4.1 Android Acquisition

- Determine what acquisition to perform - dead or live.
- Determine the type of acquisition: Physical, Logical or File System back ups
- Dead acquisition will not modify the device, but live acquisition will make modifications to the device.
- Things needed for acquisition:
    - connection between the phone and device
    - an exploit (a vulnerability which we will use to access the device to make a forensic image) to the phone, such as root access for live acquisition and different modes for dead acquisition
    - and command to image the device

For live acquisition

- exploit the device while it is booted into Android.
- Gain root access to the device
- Document and photograph (if possible) the whole process. This will make changes to the device.
- Ensure the device is charged over 50%
- Ensure debugging is enabled
- Enable Stay awake feature
- Increase screen timeout
- Remove screen lock and uncheck “Verify Apps”
- Starting with Android 6, Full Disk Encryption is enabled by default

Make sure to use “data cable” and not a “charging cable”

Backups

- ADB backup for Android 4.0 and later which requires unlocked device and USB debugging enabled; results in a tarball file. Individual apps can also be backed up.
- Other Backups can be used such as HoloBackup, which runs from a computer with the android device attached and ADB enabled.
- Helium is another option of GUI backup; it runs from an app (stores to a SD card by default) and does not require root if Helium is installed on the device and PC.

When Device is locked

- Smudge attack
- Access to the GMU account, if possible
- Brute-force is also another option, but keep in mind about failed attempts and device wipe.
- Bypass repair tools can be used such as BST dongle.
- JTAG
- Boot loader as dead exploit
- Boot the device into alternative state, use custom recovery, such as, ClockworkMod, or TWRP if available for the device.
- Use ADB to dd, but this can be risky as it may result in factory reset in some devices.

4.2 iOS Acquisition

- Find and document the details of the device: such as order number, model number, EMC number, Serial number, OS version etc.
- Determine the type of acquisition: Physical, Logical or File System back ups
- Determine iOS mode of operation: Normal Mode, Recovery Mode, or DFU Mode.
- Boot ROM vulnerability method can be used in DFU mode for iPhone 3G, 3GS, and 4; iPod Touch 2G, 3G and 4G; and iPad I models.
- Jailbreak can be used (with proper permission) to do a physical acquisition on devices without Boot ROM vulnerability. (It is recommended to do a logical acquisition before any jailbreak since it makes changes to the device)
- iOS 11 introduces SOS Mode, which requires PIN and not Biometrics, and it allows iTunes password reset.

- iOS 11.4 introduces USB restricted Mode.
- For logical and backups, variety methods can be used, such as iTunes or iCloud backups
- For Physical Image, Jailbreak method can be used
- For File systems, gray key and CAS can be used

Backups

- iTunes Backup: make sure auto syncing is disabled.
- iCloud Backup can be extracted using apple ID and password. It can take a long time.
- 2 Factor authentications may be enabled for devices with iOS 11 and later.

For Locked Devices

- Commercial Bypass such as File System Image Memory Image using Gray Key
- Cellebrite Advanced Services for File System Image
- Access to Notifications and SIRI requests can still be possible on certain devices

4.3 SIM and SD CARDS

- SIM cloning- used when a phone requires a SIM for imaging. Make sure not to use a random SIM in a phone as it will make changes to the data
- SIM cloning will isolate the phone from network and will provide entry to the device if the user has enabled SIM PIN security
- Be careful of PIN/PUK locks: after three failed attempts of PIN, PUK will appear and after 10 failed attempts of PUK, the sim will be permanently locked.
- UICC Imaging for file system and logical dumps. Communications can be done Via APDU commands and should be read outside of the phone
- SD cards and other storage medial need to be properly write-blocked before acquiring image

5. Analysis

- Read the Request and the Search Warrant to determine the scope of what needs to be analyzed.

- Check if the forensic workstation is updated with current software and hardware. Ensure that proper drivers are installed.
- Based on the scope, determine what tools to be used.
- If working on a forensic image, verify the hash with the hash provided.
- Document any third party or internally generated functions, scripts ran, or code that was used to analyze the data
- After the completion of analysis, produce a verification hash of the digital working copy
- Document the process throughout the process.
- Don't reply on only one tool. Check with multiple tools and verify manually, if possible.

5.1 Android Analysis

- For File System Backup, ADB is needed
- Android OS has many flavors (such as Apple Pie, Banana bread, Gingerbread, Nougat, KitKat, lollipop, jellybean etc.) and may need different approaches and tools to analyze
- They may have various file systems such as YAFFS, YAFFS2, RFS, EXT, F2FS
- May have SD card File Systems (FAT/VFAT)
- Different partitions may use different File systems
- Partitions of interest: cache, data, emmc, sdcard, userdata, dbdata
- Various File Types: APK, DEX, LOG, DB, JSON, XML, TXT, SQLITE
- Multi-Users supported by default in Android 5.0:
    - /user/0: Symbolic Link -> /data/data default user path
    - /user/10:/data/data for new next user
- Starting with Android 6, Full Disk Encryption is enabled by default.
- If it has a lock screen, then it must support encryption.
- If AES performance higher than 50MiB/S; MUST be encrypted
- Full Disk Encryption: AES 128-bit Key (Default: AES-CBC-ESSIV)
- Security changes starting with Android Marshmallow

Android Marshmallow:

- Fingerprint API support,
- Built-in Encryption, when device meets certain specifications (such as, lock screen)
- New Password Storage System, Gatekeeper
- Adoptable Storage

- Four Types of Boot Password: Default, Pin, Password and Gesture
- Default password is default_password, 128-bit key generated, hashed with default password and salt.
- On Password change, only 128-bit key is re-encrypted and stored.
- ISP and Chip-off acquired data will not be decrypted if the chip was encrypted. Custom Recoveries like TWRP 3.0+ versions will be needed. Ensure no handset lock option for bootup or provide passcode.
- Gatekeeper manages and verifies passwords with a hardware-backed secret key
- New storage files: Gatekeeper. Password. Key, Gatekeeper.Pattern.key
- Adoptive storage: MicroSD or OTG drives turned into encrypted internal Storage, which can be mountable in Linux. It uses 128-bit AES key and formatted in EXT4

<u>Android 7.0-7.1.1: Nougat</u>

- All 7.0 and higher devices support Direct Boot even if it is not encrypted
- File Based Encryption (FBE) encrypts file CONTENTS with AES -256 bits key length (XTS mode), and encrypts file NAMES with AES 256 bits key length (CBC-CTS mode)
- Contains 2 Separate Storage Areas:
  - CE: Credential Encrypted: Available only after credentials entered.
  - DE: Device Encrypted: Available after boot; No password
- New User directory: new _ce and _de folders for User, Misc and System
- Path changes for:
  - <u>SMS/MMS:</u> *user_de/0/com.android.providers.telephony/databases/smsmms.db*
  - <u>Call logs:</u> /data/com.android.providers.contacts/databases/calllog.db
- Encryption: Full Disk Encryption (FDE): Applies to user partition and not the entire system; On by default in Lollipop-Nougat
- Methods for cracking encryption (through Marshmallow) are available

<u>Android 10</u>

- Scoped Security is introduced and will be mandatory in iOS 11
- New permission - “read privileged phone state” for identifiers like IMEI

- <u>Artifacts of interest:</u>
  - **Bluetooth devices:** /data/com.android.bluetooth/dtopp.db
  - **Browser bookmarks:**

    /data/com.android.browser/browser.db/dbdata/databases/com.android.browser/browser.db
  - **Browser history:** */data/com.android.browser/browser.db/dbdata/databases/com.android.browser/browser.db*
  - **Browser searches:**

    /data/com.android.browser/browser.db/dbdata/databases/com.android.browser/browser.db
  - **Call history:**

    /data/com.android.providers.contacts/contacts2.db/data/com.Motorola.clur.providers.contacts/contacts2.db/data/com.sec.android.provider.logsprovider/logs.db/dbdata/databases/com.sec.android.provider.logsprovider/logs.db/data/com.android.providers.contacts/databases/calllog.db (Android 7 & up)
  - **Contacts:**

    /data/com.android.provider.contacts/contacts2.db/dbdata/databases/com.sec.android.provider.contacts/contacts2.db
  - **Media:**

    /data/com.android.providers.media/external.db/dbdata/databases/com.android.providers.media/external.db
  - **MMS and SMS:**

    /data/com.android.providers.telephony/mmssms.db/dbdata/databases/com.android.providers.telephony/mmssms.db/user_de/0/com.android.providers.telephony/databases/smsmms.db (7.0 and up)
  - **Wi-Fi hotspots:** */misc/wifi/wpa_supplicant.conf/wifi/bcm_supp.conf*
- Installed applications:
  - /system/packages. List
  - /system/packages.xml
- Wi-Fi connections:
  - /misc/wifi/wpa_supplicant.conf

        - /wifi/bsm_supp.conf
    - Wi-Fi hotspot
        - /misc/wifi/hostapd.conf
            - SSID that was transmitted by device
        - /misc/wifi/softap.conf
            - Hotspot and password configuration
    - Mobile Dialing Number: /property/persist.radio.sdma.mdn
    - ADB enabled/disabled: /property/persist.service.adb.enable
    - Registered country: /property/persist.sys.country
    - Display language: /property/persist.sys.language
    - Time zone: /property/persist.sys.timezone
    - USB services: /property/persist/usb.config
    - User accounts: /system/accounts.db
    - App usage/ last launch: /system/dpmappmgr.db
    - Gesture lock: /system/gesture.key
    - Password: /system/password.key
    - SIM information: /system/SimCard.dat
    - Sync accounts: /system/sync/accounts.xml
    - Other artifacts such as android usagestats; which tell us what file was in the foreground and background, battery status, recent images, recent tasks, snapshots are very important.
    - Third party apps, such as web browsers, chat apps, and email are valuable locations
    - Google Cloud Activity is another location to look for, if possible

5.2 **iOS Analysis**

- iOS 10.3 and up will have APFS file system. Prior 10.3 will have HFSX or HFS+ File System
- iOS 4 – iOS 9 backup structure will have SHA1 named Files, Plist Files, MBDB etc.: iOS 10+ backups have new folder structure. Plists still exist, but DB replaces MBDB
- Manifest.db table contains FileIDs (sha1 hash values); Domain + relative paths and metadata in Binary Plist about File
- Sensitive Data such as HomeKit, HealthKit, keychain (key changes)

- Data in Files Column is binary formatted Plist and stores User ID, path, last modified time, protection class and public Encryption key
- iOS 10 introduces iTunes prompt for device back up
- iOS5 introduces encryption and will not give unallocated data
- iTunes backups – iOS 4 introduces encryption
- AFC
- Other services
    - com.apple.mobile.house_arrest (till iOS 8)
    - com.apple.mobile.file_relay (till iOS 8)
- AFC2 - Requires jailbreak
- Multiple encryption changes in iOS 10.0, 10.1 and 10.2
    - iOS 10.0 was flawed and allowed for millions of attempts per second
    - 10.1 keybag still in manifest. plist, but key removed from manifest.db
    - 10.2 introduces Heavy Encryption; requires 2 salting steps and more iterations
    - 10.3 Manifest.db is encrypted and contains public encryption key, manifest.plist is altered –but contains key+salt

<u>iOS 11</u>

- New pairing restrictions require physical passcode
- TouchID/ FaceID can unlock phone but not connect to device.
- Backup encryption still using iOS 10.2+ recursive iterations.
- iOS 11 adds option to remove previously set backup encryption
- Emergency SOS Mode introduced
    - Press power rapidly 5 times OR
    - Side Button Plus Volume Button
- Disables touch ID
- Send message to designated contacts
- This is a problem when you can compel a fingerprint but not a password
- Pairing a device must use PIN and not touch ID
- Pairing records still available
- iCloud backups don’t require a PIN; 2 Factor Authentications may be on the device and may be answered my biometrics
- iOS 2FA
    - It is not just an SMS/MMS

    - Device must be a “Trusted Device” through iCloud
    - Presents a Location from where device is being accessed
    - 6 Digit PIN is hidden behind lock screen
    - Can unlock with biometrics and read the PIN
- iOS 11.4.1 introduces USB Restricted mode, only charge and no data pass through the port. No imaging can be done
- SOS enables Restricted Mode; iOS device doesn’t register to Windows OS or to Registry values. USIS is not trackable via registry
- SIRI requests:
    - Calendar -shows your last 25 requests
    - Last Call
    - What’s my name –will show user’s contact card
    - Call “name” - will give the choice of names, but will ask to unlock the phone
    - How long till home? –will give the distance to home so the address (need to unlock for work!) can be found.
- iOS 11 introduces HEIF (High Efficiency Image Format) for photos, and HEVC(High Efficiency Video Codec) for videos:
    - Provides better compression than JPEG and H.264
    - uses less storage space.
    - Users can select format in Coding format
    - Only encoded by Apple devices with A10 chips or later (iPhone 7 and newer)
    - It is an option to use MTP transfer mode
    - If Automatic is selected – file will be converted to a compatible format when transferred
    - HEIC will have a matching MOV file with it (by filename)
    - User can change display thumbnail by selecting one of the frames from the .MOV file; stored as ‘FullSizeRender.jpg’ and is considered a Mutation
    - MOV can contain audio if sound not muted
    - Once thumbnail is edited or live photo has mutation applied, .MOV also gets a ‘FullSizeRender.mov’ variant.
- Notifications: Cleared Notifications are no longer stored
- Do Not Disturb While Driving – Preference Entry; Contains last 5 times user drove car
    - Location – CARDNDMiniRecentRideHistory
- Photo Locations :

    - */private/var/mobile/Media/DCIM/100APPLE/*
    - */private/var/mobile/Media/PhotoData/Mutations/DCIM/100APPLE/[IMG NAME]/Adjustments/*
    - *Adjustments.plist*
    - *FullSizeRender.jpg / FullSizeRender.mov*
    - *private/var/mobile/Media/Thumbnails/V2/DCIM/100APPLE/[IMG NAME]*
    - *5005.jpg (even if the image is HEIC)*
- Native Files App: Allows users to browse iCloud and local files
- Calendar events: Can contain additional info such as invites and attendees
- Addressbook.sqlite - contains new fields in the ABPersontable
- New vision framework - support for detecting faces, bar codes, image horizon, rectangular regions
- Native camera app support for bar Codes and QR Codes
- Safari History: device displays 30 days' worth of data (may be kept longer than 30 days if it's a repeat visit to URL) regardless of Safari in iCloud.
    - Database contains deleted entries, valuable carved history
- iMessage account in the cloud (introduced to iOS11.4) will give access to messages, which are separate from backups.
    - Not all messages will stay caught on the device if enabled.
    - Not all messages will be cached to iCloud Backups

<u>iOS 12</u>

- Similar in structure to iOS 11
- Screen Time - Tracks app usage, when device most used, notifications; meant to be parental control.
- Group FaceTime Calls - new table in database: ZHANDLE, Z_REMOTEPARTICIPANTHANDLES; Similar to iMessage / SMS address tracking
- Call Logs - new table in database: ZHANDLE, Z_REMOTEPARTICIPANTHANDLES

<u>iOS 13</u>

- Safari and Call logs only in Encrypted Backups!
- .ktx files are used for snapshots of minimized apps
- iPads now run iPadOS starting with 13, with downloadable files from Safari
- More artifacts and locations of interest

- App permissions - */Library/TCC/TCC.db*
- Installed Applications:
    - */private/var/mobile/Applications*
    - */private/var/mobile/Containters/Data/Application*
- Manifest Plist
    - */BackupService/9108a1e75cdbd7700bd4411609ee5cfa6e71013e/Snapshot/Manifest.plist*
    - */private/var/root/Library/Caches/Backup/Manifest.plist*
- Launch services Daemon (LSD)
    - */private/var/db/lsd/com.apple.lsdidentifiers.plist*
- File System Structure
    - App Data
        - Documents
        - Library
            - caches
        - Tmp
    - Var
- With a Full File System image – many valuable information can be found:
    - Cached Locations
    - Safari
    - Network Usage –Connection Data, Application Data
    - Screen Time Synced Applications
    - Apple Maps Trips, Significant locations and visit history, Power Log
    - Knowledge artifacts, such as, Application focus, Device Lock states, Application activities, Media history etc.
    - Third party apps

5.3 SIM CARD Analysis

- Subscriber Identity Modules (SIM) cards contain sensitive information about the subscriber. Data such as contact lists and messages that are valuable to the examiner:
    - ICCID: The integrated circuit card identification is a unique numeric identifier for the SIM that can be up to 20 digits long. It consists of an industry identifier prefix, followed by a country code, an issuer identifier number, and an individual account identification number.

- IMSI: The international mobile subscriber identity
  is a unique 15-digit number provided to the subscriber. It has a similar structure to ICCID and consists of the MCC, MNC, and MSIN.
- MSISDN: The Mobile Station International Subscriber Directory Number is intended to convey the telephone number assigned to the subscriber for receiving calls on the phone.
- Abbreviated dialing numbers (ADN): Any number and name dialed by the subscriber is saved by the ADN EF. The type of number and numbering plan identification is also maintained under this
- Fixed dialing numbers (FDN): The FDN EF works similar to the ADN because it involves contact numbers and names. With this function, the user doesn't have to dial numbers; by pressing any number pad of the phone, he can access the contact number.
- Last number dialed (LND): The LND EF contains the number most recently dialed by the subscriber. The number and name associated with that number is stored in this entry
- Other Valuable information can be extracted from SIM Analysis such as:
  - Service provider name (SPN)
  - Mobile country code (MCC)
  - Mobile network code (MNC)
  - Mobile subscriber identification number (MSIN)
  - Mobile station international subscriber directory number (MSISDN)
  - Abbreviated dialing numbers (ADN)
  - Last dialed numbers (LDN)
  - Short message service (SMS)
  - Language preference (LP)
  - Card holder verification (CHV1 and CHV2)
  - Ciphering key (Kc)
  - Ciphering key sequence number
  - Emergency call code
  - Fixed dialing numbers (FDN)
  - Local area identity (LAI)
  - Own dialing number
  - Temporary mobile subscriber identity (TMSI)

- Routing area identifier (RIA) network code
- Service dialing numbers (SDNs)

5.4 SDCARD/emmc Analysis

- SD cards, MicroSD cards or OTG drives can be used as Adopted internal storage to save everything in the device only (including apps and photos); Requires formatting that prevents it from working with other devices
- It can also be used as portable storage to store photos and other media between devices
- Folders of interest in portable storage:
    - Android/data
    - DCIM
    - Download
    - Pictures
    - ScreenCapture
    - Various app folders

5. 5 When Device is off or damaged

If the device is not supported by physical, file system or logical acquisitions, PIN/Password protected and cannot be examined without a password/PIN, or is damaged, consider photography, manual transcription and advanced mobile device techniques such as JTAG/Chip Off

6. Documentation

Documentation may vary based on scenarios or cases. However, at a minimum it must include:

6.1 Request:

The examination request- It should clearly define what is requested of the examiner and the examination)

6.2 Chain of Custody (CoC):

At a minimum, COC must include:

- a clear description of the evidence provided including location found, evidence description, device state (locked/unlocked, on/off), processing need, owner name, case number, person collecting, custody chain (person/date/time)
- and a detailed history of the transfer of the evidence in a chronological order including seizure and beyond.
- Take photographs from all sides without removing anything. Ensure that it maintains integrity of the evidence
- COC applies to both physical device and the forensic image created from the device. The forensic image becomes evidence.

6.3 Notes

At a minimum, notes should include:

- Examiner notes regarding the case
- Steps taken from beginning to the end of the examination with date and timestamps and sufficient detail that will allow another examiner with similar level of expertise and experience to follow through the process and get to similar findings.
- Signature and initial of the examiner in each page of the notes, if multiple examiners are involved then add their initial to respective sections.
- All notes should be contemporaneous, complete and discoverable.
- Examiners should avoid making generalizations and giving opinions.
- Notes should contain only facts about procedures, methods and outcomes
- It should be a guide for creating the final report.

6.4 Reporting

Reporting will vary based on the lab or case. There may be more than one report depending on the organization/case. But a at a minimum report must contain the following:

- Name and logo (if any) of the reporting organization
- Case identifier
- Name and title of the Requesting Agent or submitter
- Date and time of the receipt.
- List of the items provided for the examination – for example the evidence, additional media, documentation (warrant, consent)
- Request Description: what was asked of the examiner

- Date and time of the report
- Name and signature of the examiner, and initials in every page of the report
- Screenshots and/or other graphics to explain findings or artifacts
- References Page to explain technical terms

Reporting should be in the following format:

- Executive Summary
- Complete seizure and applicable authority
- Collection (forensic acquisition, network isolation, tools used with version number)
- Analysis – Tools used, Data meaning
- Correlating the data from sources and examiner comments
- Conclusions – Results, Items derived
- Exhibits- Screenshots and/or other graphics to explain findings or artifacts

Works Cited

Ayers, Rick, Sam Brothers, and Wayne Jansen. "Guidelines on Mobile Device Forensics." National Institute of Standards and Technology, May 2014. https://doi.org/10.6028/NIST.SP.800-101r1.

"DEUSOP02 - Mobile Device Acquisition.Pdf." Accessed May 7, 2020. https://dfs.dc.gov/sites/default/files/dc/sites/dfs/publication/attachments/DEUSOP02%20-%20Mobile%20Device%20Acquisition.pdf.

Infosec Resources. "SIM Card Forensics: An Introduction," November 19, 2013. https://resources.infosecinstitute.com/sim-card-forensics-introduction/.

"SWGDE_Model_SOP_for_Computer_Forensics_091312.Pdf." Accessed May 7, 2020. https://www.irisinvestigations.com/wp-content/uploads/2016/12/ToolBox/04-ISO%20QUALITY%20MANAGEMENT%20SYSTEM/SWGDE_Model_SOP_for_Computer_Forensics_091312.pdf.

The Cyber Security Place. "Introduction to Mobile Forensics - Android OS," March 22, 2017. https://thecybersecurityplace.com/introduction-mobile-forensics-android-os/.